Radial Basis Function (RBF) Kernel

RBF Kernel

$e^{-\gamma||a-b||^2}$

$e^{-1/2||a-b||^2} = e^{-1/2(a^Ta + b^Tb - 2a^Tb)} = e^{-a^2/2} \cdot e^{-b^2/2} \cdot e^{aTb}$

Taylor expansion of e^{aTb}:

$1 + a^{Tb}/\sqrt{1}! + (a^{Tb})^2/\sqrt{2}! + (a^{Tb})^3/\sqrt{3}! + (a^{Tb})^4/\sqrt{4}! + \ldots \infty$

$\varphi(a)^T = e^{-a^2/2}\,[1, a_1/\sqrt{1}!, \ldots]$

$\varphi(b) = e^{-b^2/2}\,[1, b_1/\sqrt{1}!, \ldots]^T$

$\varphi(a)^T\varphi(b) = e^{-a^2/2}\, e^{-b^2/2}\, [1 + a^{Tb}/\sqrt{1}! + (a^{Tb})^2/\sqrt{2}! + \ldots] = e^{-a^2/2}\, e^{-b^2/2}\, e^{aTb}$

$= e^{-||a-b||^2/2}$

b series:

1

$b/\sqrt{1}!$

$b^2/\sqrt{2}!$

$b^3/\sqrt{3}!$

$b^4/\sqrt{4}!$

...

∞

$e^{-b^2/2}$